

MGMT 5900 Big Data Analytics

Data Ninja 7 Team – Assignment #1

Team Members:

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Agenda

- ❖ Problem Statement
 - ❖ Key Findings
 - ❖ Integrated Insights: Why Profitability is Lagging
 - ❖ Strategic Imperatives for Profitability Optimization
 - ❖ Realizing a More Profitable Future & Next Steps
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Problem Statement:

Superstore Co. faces significant profitability disparities across product categories and geographic regions, with alarming declines in key areas despite revenue growth, indicating structural issues that erode overall profitability.

Key Findings:

Overall Performance: Strong operational foundations, but significant disparities exist.

Product Performance:

- ♦ **Strengths:** Technology & Office Supplies are primary revenue and profit drivers.
- ♦ **Challenge:** Furniture is a critical underperformer (3.86% profit margin, 56.63% profit decline).
- ♦ **Loss-Making Products:** Identified specific items like Cubify CubeX 3D Printers, Lexmark printers, and conference tables.

Customer Behavior: Discounts exceeding 30% create negative profits. Higher quantities per order correlate with increased profit.

Regional Performance: West and East are leaders; Central and South offer growth opportunities.

Integrated Insights: Why Profitability is Lagging

- **The Furniture Drain:** The Furniture category's severe **56.63% profit decline** (2021-2022), despite revenue growth, indicates fundamental issues with its cost, pricing, or operational efficiency, significantly dragging overall performance.
- **Discounting Dilemma:** Aggressive discounts, particularly **exceeding 30%**, directly erode profitability and often fail to generate proportional sales volume, highlighting a need for a more strategic pricing approach.
- **Product Portfolio Imbalance:** A significant portion of the catalog, including **Tables, Bookcases, Supplies, Binders**, and specific **printers**, consistently operates at negative margins, diluting the strong profits from "Star" products like **Copiers and Phones**.
- **Regional Disparities:** Lower performance in Central and South regions, compared to the strong West and East, suggests **missed opportunities** due to suboptimal product mix or untargeted strategies.
- **Quantity Correlation:** Selling **higher quantities per order** directly correlates with increased profitability, indicating that larger, multi-item transactions are more efficient and profitable.

Strategic Imperatives for Profitability Optimization

Furniture Category Restructuring:

- ♦ **Action:** In-depth cost analysis; phased discontinuation of unprofitable items.
- ♦ **Goal:** Optimize product mix for profitable furniture offerings.

Discounting Policy Optimization:

- ♦ **Action:** Immediately eliminate discounts >30%.
- ♦ **Strategy:** Develop data-driven campaigns balancing sales volume with margin preservation.

High-Value Product Focus:

- ♦ **Action:** Heavily invest in and promote "Star" products (e.g., Copiers, Binding Machines).
- ♦ **Goal:** Maximize returns from proven profitable products.

Additional Strategic Focus Areas:

- ♦ Enhance Regional Strategies (West/East reinforcement, Central/South growth).
- ♦ Incentivize Larger Order Quantities.
- ♦ Proactive Management of Unprofitable Products/Orders.

Realizing a More Profitable Future & Next Steps

Expected Impact:

- **Enhanced Profitability:** Significant improvement by eliminating loss-making activities.
- **Streamlined Operations:** Reduced complexities and costs.
- **Sustainable Growth:** Healthier business model focused on high-margin products.
- **Improved Decision-Making:** Data-driven insights for proactive strategies.

Next Steps:

- Form cross-functional teams for implementation.
- Establish clear KPIs and monitoring mechanisms.
- Re-evaluate product portfolio and discounting policies quarterly.